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RESEARCH ARTICLE

# Anchoring bias in large language models: an experimental study

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## Abstract

Large Language Models (LLMs) like GPT-4 and Gemini have significantly advanced artificial intelligence by enabling machines to generate and comprehend human-like text. Despite their impressive capabilities, LLMs are not without limitations. They exhibit various biases. Although much research has explored demographic biases, cognitive biases in LLM have not been equally studied. This study delves into anchoring bias, a cognitive bias in which initial information disproportionately influences judgment. Using an experimental data set, we examine how anchoring bias manifests in LLM and verify the effectiveness of various mitigation strategies. Our findings highlight the sensitivity of LLM responses to biased prompts. At the same time, our experiments show that to mitigate anchoring bias, one needs to collect information from comprehensive angles to prevent the LLMs from being anchored to individual pieces of information, as simple algorithms such as Chain-of-Thought, Thoughts of Principles, Ignoring Anchor Hints, and Reflection are insufficient.

**Keywords** Large language model · LLM · Anchoring bias · Cognitive bias · Mitigation strategy

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# 1 Introduction

Nowadays, large language models (LLMs) such as GPT4 and Gemini have revolutionized the field of artificial intelligence by enabling machines to understand and generate human-like text. Furthermore, they have also shown remarkable performance in many common sense reasoning tasks. Many researchers and industry professionals have started to apply LLMs to decision-making processes to support their day-to-day business. For example, in addition to being used to generate articles, stories, and even code, LLMs are also used as chatbots and virtual assistants for instant customer service support, to help with medical research, diagnostics, and patient communication in healthcare [1, 2], and many other decision-making scenarios. Despite their impressive capabilities, LLMs are not without limitations. In fact, a set of studies have shown that LLMs have social or stereotype bias problems [3], and these biases can have negative impacts. Quite a lot of attention has been paid to better understand and eliminate the negative impact of demographic bias in LLMs, by both the research community and industry.

Unlike demographic biases of LLMs based on sensitive characteristics such as race and gender that have been widely studied by researchers [4], cognitive biases of LLMs have not received much attention yet. There are very few research papers focusing on cognitive biases in LLMs. Cognitive biases in LLMs can affect the fairness and accuracy of their outputs [5, 6]. It is especially important when we use LLMs in decision-making tasks, as they may lead to unreasonable recommendations. As LLMs become more and more popular in decision making scenarios, there is a high demand for a comprehensive understanding of LLMs' cognitive biases and for practitioners to be cautious about them to avoid irrational decision-making output.

This paper aims to focus on a specific type of cognitive biases—anchoring bias, and study its manifestation in LLMs and possible mitigation strategies. Unlike [7], where only a few financial questions are examined, we conducted a more comprehensive and quantitative study on the anchoring bias of LLMs based on the experimental dataset designed by Taha Yasseri [8]. By changing the magnitude of bias hints, we found that the answers of LLMs are sensitive to the biased hints. We further conducted a comparative study on several mitigation strategies. Our study shows that simple algorithms such as Chain-of-Thought, Thoughts of Principles, Ignoring Anchor Hints, and Reflection are insufficient to mitigate anchoring bias. One needs to collect hints from comprehensive angles to prevent the LLMs from being anchored to individual pieces of information.

# 2 Related work

## 2.1 Cognitive bias

Cognitive bias refers to systematic mental patterns that influence our thinking and decision-making, leading us to process information in a selective and subjective manner, often resulting in inaccurate or irrational judgments [9]. It is a systematic error in thinking that affects how we process information, perceive the environment, and make decisions. It can lead to irrational thoughts or judgments and is often based on our perceptions, memories, or individual and societal beliefs. Cognitive bias is often an unconscious and automatic process—a result of the brain's attempt to simplify information processing, which is designed to make decision-making quicker and more efficient. Individuals are often unaware of their attitudes and behaviors resulting from cognitive bias [9]. Unlike demographic bias, where behavior occurs due to social and cultural influences, cognitive bias lies in the information processing mechanisms of the human decision-making process, often influenced by the setup of the task or emotional factors.

There are many cognitive bias patterns that have been studied in Psychology. For example, confirmation bias, hindsight bias, the mere exposure effect, self-serving bias, the base rate fallacy, anchoring bias, availability bias, the framing effect, inattentional blindness, and the ecological fallacy are some of the most common examples of cognitive bias [9].

In this study, we mainly focus on anchoring bias [10]. Anchoring bias means that our judgments are often significantly influenced by the first piece of information encountered. For instance, once an anchor (i.e., the first piece of information encountered) is established from a previous message, people often insufficiently adjust away from it to arrive at their final answer, making their final guess or decision closer to the anchor than it otherwise would have been.

### 2.2 Cognitive biases in LLMs

LLMs are trained by a next-token prediction task based on the text data collected from the Internet. Therefore, LLMs are susceptible to various biases because the data they are trained on inherently includes biases. Additionally, the algorithms used to train these models can exacerbate or introduce new biases depending on their design and the assumptions they make during learning. The interaction between users and the model can further entrench these biases, as frequent user inputs may reinforce certain patterns that the model learns to replicate. These biases can potentially manifest in several ways at the LLM inference time, impacting the reliability and fairness of the model's outputs.

A few existing research studies have shown that LLMs can show cognitive biases in many different decision-making processes [3, 11, 12]. For example, Chen et al. [12] pointed out that LLM's judgments are influenced by threshold priming biases. In [11], the authors detected several cognitive biases (confirmation bias, anchoring bias, status quo bias, framing bias, primacy bias, group attribution bias, etc.) in LLMs by using a framework called BIASBUSTER. Samuel et al. [6] assessed cognitive biases in LLMs used for medical tasks by using a benchmark called BiasMedQA, and showed that LLMs are vulnerable to cognitive biases, while *GPT4* is demonstrating significant resistance to bias. However, the research of [13, 14] showed that although GPT models are frequently subject to cognitive biases, the way these biases are displayed does not reflect that shown by humans. Ross et al. [15] also evaluate the cognitive biases of LLMs quantitively using a utility theory in the financial domain and found that the financial behavior of current LLMs is neither entirely human-like nor entirely homo-economicus-like.

In this paper, we focus on the anchoring bias of LLMs including its behavior and mitigation strategies. There are a few existing works focusing on anchoring bias of LLMs. Li et al. [16] studied LLMs' anchoring bias on answering multiple-choice questions (MCQs). To mitigate it, the authors identified the Multi-Layer Perceptron (MLP) and attention neural vectors that are responsible for this bias and updated these vectors to neutralize the bias to the anchored choice 'A'. However, their approach can only be applied to limited scenarios i.e., MCQs. The most related work to our research in this paper is from [7] where the authors tested whether LLMs are subject to anchoring bias and studied two naïve mitigation prompting strategies. They found that two naïve mitigation prompting strategies including Chain of Thought (CoT) and an "ignore previous" strategy (using a prompt: "Ignore our previous conversation but remember your role") cannot consistently reduce biases for different LLMs. However, this study is only based on 5 questions in the financial domain. Different from these existing ones, our study includes 62 questions from different domains and gives a quantitative analysis about the effects of anchoring bias with different hints.

## 3 Our approach

In this study, we conduct a systematic experiment to assess anchoring bias of LLMs through an in depth quantitative analysis and explore potential mitigation strategies based on the dataset from [8].

### 3.1 Data set

In order to systematically evaluate the anchoring effects of LMMs, we use a dataset from [8] which was originally collected for a quantitative analysis of human anchoring biases. To make the data representative and closer to everyday life, the data is collected from the community of the "Play the Future" (PTF) game, where users make digital predictions about the outcomes of various economic, social, sports, entertainment, and other events. Figure 1 gives three typical examples. Each sample contains a question $Q$ requiring a numerical answer, and three hint messages $H_1$, $H_2$, and $H_3$. For example, "Over 250 craft, import, and domestic beers will be presented at Ottawa's Beer Fest this weekend. The entry fee covers 10 drink tickets and a sampling cup. What will the weather be like at 1 PM in Ottawa, Ontario, on Sunday?" is the question of the second sample in Fig. 1, which is asking for a temperature value. The first hint $H_1$ of this sample ("Temperature last Sunday at 1 PM: 68°F") is a fact that usually provides a useful reference information for answering the numerical question $Q$ of this sample. $H_2$ and $H_3$ are two different anchoring hints that may potentially trigger the LLMs to give biased answers about the question topic. In this case, $H_2$ ("Tom from PTF's prediction: 60 F") gives a lowvalue anchor (i.e. 60 F) and $H_3$ gives a highvalue anchor (i.e. 76 F). The whole dataset contains 62 samples (and their corresponding hints) covering multiple aspects of daily life, including some natural, economic, social, sports, and entertainment events, such as weather, stock prices, and flight times.

| Question Type | Event name | Question Example | Hint H1 | Hint H2 | Hint H3 |
|---|---|---|---|---|---|
| Standard | Elon Musk Birthday | Elon Musk is celebrating his birthday this week: he was born on June 28 in 1971. He is a forward -thinking businessman known for his companies SpaceX, Tesla, Inc. and his latest project, Hyperloop, a high - speed transportation system. In USD, what will Tesla's (TSLA) stok price be at the Nasdaq Stock Market next Tuesday (July 3) at 1 PM? | TSLA's stock price last Tuesday at 1 PM: 334.34 USD | TSLA's stock price low last Tuesday: 326.00 USD | TSLA's stock price high last Tuesday: 343.55 USD |
| PTF-prediction | Ottawa Craft and Beer Festival | Over 250 craft, import, and domestic beers will be presented at Ottawa's Beer Fest this weekend. The entry fee covers 10 drink tickets and a sampling cup. What will the weather be like at 1 PM in Ottawa, Ontario? | Temperature last Sunday at 1 PM: 68 F | Tom from PTF's prediction: 60 F | Sally from PTF's prediction: 76 F |
| Irrelevant anchor in control group | Friday the 13th | The weekend is almost there! This week, Friday is not just any Friday but Friday the 13th. Did you know that the 13th of a month falls at least once on a Friday every year but not more than three times? How many upvotes will the last post on Friday in the subreddit r/IsTodayFridayThe13th receive within 24 hours? | Number of subscribers to the subreddit: 25.200 | Lowest number of upvotes last 10 posts: 510 | Highest number of upvotes last 10 posts: 628 |

**Fig. 1** Question Examples

There are three different types of anchoring hints included in the dataset to assess different types of anchoring situations: fact anchoring, "expert-opinion" anchoring, irrelevant anchoring. Different types of anchoring questions may trigger different bias behaviors of LLMs. These different types of questions and hints allow us to evaluate how LLMs predictions are influenced by these different anchoring facts.

Figure 1 shows three typical question samples. In the first row of Fig. 1, the hints of $H_2$ and $H_3$ demonstrate a fact anchoring situation, where the $H_2$ and $H_3$ anchoring hints are factual descriptions of the highest and lowest stock prices last Tuesday. Although neither hint can directly provide the answer to the question (next Tuesday's stock price at 1 PM), seeing different hints ( $H_2$ and $H_3$ ) may lead to different LLM answers. In other words, these two hints may cause the models to show anchoring effects. In the second row of Fig. 1, the $H_2$ and $H_3$ hints provide two different expert opinions, which we can use to evaluate the impact of expert opinions on LLM predictions. In the dataset of [8], there are also 8 cases whose $H_1$ hints do not directly relevant to the answers of the corresponding questions. For example, in the third row of Fig. 1, the question is asking the number of upvotes to a post, but it's $H_1$ hint of "Number of subscribers to the subreddit: 25.200" is not directly relevant to the question. We try to use these cases to test the robustness of bias effect (refer to Fig. 4).

### 3.2 Experiment design

To evaluate the anchoring effects of LLMs, we also need to prompt a LLM $M$ to answer question $Q$ under different anchoring hints. For each question $Q$, we design three experiments: control $C$, treatment $A$, and treatment $B$. In a control experiment, $Q$'s first hint $H_1$ is shown to LLM $M$ along with the question $Q$ as a part of the prompt string. In treatment experiment $A$, besides the question $Q$, the anchoring hints $H_1$ and $H_2$ are both shown to LLM $M$. Similarly, hints $H_1$ and $H_3$ are shown to $M$ in treatment experiment $B$. Through systematically manipulating the visibility of the anchoring hints $H_2$ and $H_3$ to LLM $M$, we can examine the influence of anchoring hints on the answers from LLM $M$.

The results generated by LLM are often sensitive to the prompt that guides the LLM to answer our questions. In this study, we follow the CO-STAR format [17] to compose our prompt and guide the LLM towards the desired outcome, which often consists of Context, Objective (i.e., the description of a task), Style & Tone, Audience, and Response (i.e., output format). Because our task is relatively simple, it only needs LLMs to output a numerical value rather than a long passage. So, we ignore the Style & Tone and Audience parts because these parts are mainly used to specify the desired writing style and emotional tone for your LLM's response. Our prompt mainly consists of a context for background details including role playing, an objective of task description, and a specification of output format. Our basic prompt template is presented in Appendix A. As the provided hints often do not give enough information for an accurate answer to the question, we found that, if we simply ask a strong LLM (e.g., GPT4) to predict a value to answer our questions, the model sometimes directly refuses to give any answers. Instead, it replies to us with "no enough information for prediction". In the prompt template, we ask the LLM to generate an educated guess to answer our question rather than a prediction, which can mitigate the refuse-answering issue. We ask the LLM to generate an answer with a numerical value and its unit using a JSON like format. In our experiments, all Control and Treatments $A$ & $B$ share the same prompt template through replacing the placeholders of "user_question", "hint_1", and "hint_2" at runtime. For each question $Q$, we replace “user_question”, “hint_1” by $Q$ and $H_1$. In a Control experiment, “hint_2” is empty, and it is replaced by $H_2$ and $H_3$ respectively for different Treatment experiments.

In our study, we evaluate anchoring bias on several series of LLMs from different vendors:

1. *GPT4o*, *GPT4*, *GPT3.5 Turbo*, and *GPT-o3* models from OpenAI.
2. *DeepSeek-R1-QWen-32B* model from DeepSeek.
3. *Gemini2.5-flash* and *Gemini2.5-flash-lite* models from Google.
4. *Claud3-haiku*, *Claud3.5-haiku* from Anthropic.

Here, *DeepSeek-R1-QWen-32B, GPT-o3, Gemini2.5-flash, and Gemini2.5-flash-lite* are Reasoning models that are especially trained for deep thinking capability.

For each question, we ask every LLM model using a prompt composed based on the prompt template, collect, and parse the answer from the LLM model to extract a numerical result. In the answer parsing step, we convert datetime values in the format of “HH:MM” or “MM:SS” to a decimal value in the corresponding minimal unit, because decimal values are easy to be used in statistical analysis. For example, ’2:00’ with the format of “HH:MM” will be converted to a decimal value of ’120’ with a unit of ’minute’. Similarly, “2:00 of MM:SS” will be converted to ’120’ with a unit of ’seconds’. For each question, we run it 30 times in every language model to obtain enough data for statistical analysis. In the experiments, we set the temperature value as 0.8.

## 4 Result analysis

### 4.1 The effect of hint 1

Figure 2 shows the standard variation values of responses from different LLM models (i.e., *GPT4*, *GPT4o*, and *GPT3.5*) on Control experiments. From the figure, we found that many LLM responses of strong models (*GPT4* and *GPT4o*) only have very small variations although a relatively large temperature (i.e. 0.8) is used in the experiments. For example, in 78 experiments (out of 108 total experiments) from *GPT4* and *GPT4o*, the standard variation values are almost zero, which means the LLMs precisely repeat the same numerical answer 30 times. At the same time, for a weaker model *GPT3.5*, only one fourth of experiments have zero variation values. A reasonable explanation for this is that a weak model (i.e., *GPT3.5*) is often not confident about its output.

As mentioned in Sect. 3, there are eight samples in the dataset, where each sample’s $H_1$ is irrelevant to its question. We can use these samples to check whether the models only blindly copy the number from hint $H_1$ as their outputs without understanding the user question and the content of hint $H_1$, Fig. 3 shows the standard variation values of the responses from all LLMs. We can see that all these 8 cases have large variation values on both *GPT4* and *GPT4o*, which are quite different from other samples in Fig. 2. It means that LLMs do answer the questions based on the content

| question_id | GPT4_std | GPT-4o_std | GPT35_std | question_id | GPT4_std | GPT-4o_std | GPT35_std |
|---|---|---|---|---|---|---|---|
| 1 | 54.77 | 106.60 | 105.40 | 34 | 0.00 | 0.00 | 0.00 |
| 2 | 0.00 | 0.98 | 2.28 | 38 | 0.00 | 0.00 | 0.00 |
| 3 | 0.00 | 0.00 | 0.00 | 39 | 0.00 | 0.00 | 0.00 |
| 4 | 0.00 | 0.00 | 2.49 | 40 | 10.95 | 14.27 | 66.44 |
| 17 | 0.00 | 0.00 | 0.08 | 41 | 18.26 | 10.81 | 1937.29 |
| 18 | 0.00 | 0.00 | 0.00 | 43 | 0.00 | 1.01 | 641.43 |
| 19 | 0.00 | 0.00 | 0.00 | 44 | 0.00 | 8.71 | 159.15 |
| 20 | 0.00 | 0.00 | 3.24 | 45 | 0.00 | 0.00 | 0.00 |
| 21 | 0.00 | 0.00 | 7.13 | 47 | 0.00 | 0.00 | 0.23 |
| 22 | 66.45 | 94.32 | 104.76 | 49 | 0.00 | 3158.93 | 1853574.53 |
| 23 | 0.00 | 0.00 | 0.00 | 50 | 0.00 | 0.00 | 5356.36 |
| 24 | 0.00 | 0.03 | 0.00 | 51 | 40.32 | 25.96 | 2507.03 |
| 25 | 0.00 | 0.00 | 0.00 | 52 | 0.00 | 6.13 | 11.73 |
| 26 | 10.41 | 12.59 | 21.54 | 53 | 0.00 | 0.00 | 0.59 |
| 27 | 0.00 | 0.00 | 0.00 | 56 | 0.00 | 0.00 | 1.46 |
| 28 | 0.00 | 0.00 | 0.11 | 57 | 0.00 | 0.00 | 38.39 |
| 29 | 0.00 | 0.00 | 4608.80 | 58 | 7.90 | 72.97 | 34.35 |
| 30 | 0.00 | 0.00 | 9.89 | 59 | 0.00 | 0.00 | 2.64 |
| 31 | 0.00 | 0.00 | 14.85 | 60 | 0.00 | 0.00 | 0.00 |
| 32 | 0.00 | 0.00 | 14258146.68 | 61 | 11.44 | 71.95 | 35.33 |
| 33 | 0.00 | 0.19 | 0.33 | 62 | 0.00 | 0.00 | 0.00 |
| 5 | 0.00 | 0.00 | 0.98 | 11 | 0.00 | 0.00 | 0.18 |
| 6 | 0.00 | 77039.18 | 1507912.66 | 12 | 0.00 | 0.00 | 682.39 |
| 7 | 0.00 | 0.00 | 0.00 | 13 | 0.00 | 0.19 | 3.08 |
| 8 | 0.00 | 0.00 | 732.45 | 14 | 0.17 | 0.25 | 1.73 |
| 9 | 0.00 | 1214746.63 | 2397175.47 | 15 | 0.00 | 0.00 | 2.36 |
| 10 | 0.14 | 0.07 | 0.79 | 16 | 0.00 | 0.00 | 0.00 |

**Fig. 2** Standard deviations of answers from different GPT models given fact hint $H_1$

| question_id | GPT4_std | GPT-4o_std | GPT35_std | question_id | GPT4_std | GPT-4o_std | GPT35_std |
|---|---|---|---|---|---|---|---|
| 37 | 40.55 | 6.01 | 4.40 | 46 | 72.17 | 3318.77 | 4516.39 |
| 36 | 281.85 | 145.44 | 26690.82 | 48 | 8200.44 | 2701.20 | 10370.33 |
| 35 | 165.02 | 1690.82 | 3584.87 | 54 | 36.43 | 218.72 | 13930377.47 |
| 42 | 16.14 | 54.36 | 31.17 | 55 | 3139.38 | 4634.64 | 8364.58 |

**Fig. 3** Standard deviations of answers from different GPT models given irrelevant hints

understanding of hint $H_1$, and they cannot give confident responses when they are given an irrelevant hint $H_1$.

## 4.2 Anchoring effect evaluation

To evaluate the anchoring bias effect, we need to check whether the answers of LLMs are significantly influenced by hint $H_2$ or $H_3$ in the experiments. If anchoring effect exists, we are able to expect those answers with a high hint figure to obtain higher answer values than those who were provided with low hint figures. In other words, if the answers of a LLM under $H_2$ and $H_3$ are significantly different and the sign of the difference between them is the same as that of the difference between the original $H_2$ and $H_3$ values, we can claim that the answers of the LLM are biased by the anchors $H_2$ and $H_3$.

In our experiments, for each question, we obtain two value sets from every language model: one set contains 30 answers for Treatment $A$ (i.e. answer set $A$), and the other contains 30 for Treatment $B$ (i.e. answer set $B$). We evaluate the bias effect by estimating the average difference between two underlined distributions of answer

| question_id | Hint Difference | diff | p_value | diff_4o | p_value_4o | diff_35 | p_value_35 |
|---|---|---|---|---|---|---|---|
| 1 | 600 | 123.33 | 2.69E-04 | 316.63 | 1.56E-08 | 398.68 | 9.13E-14 |
| 2 | -2.09 | -0.15 | 8.15E-02 | -0.77 | 1.56E-07 | -5.49 | 1.05E-01 |
| 3 | -0:16 | -0:00 | 1.00E+00 | -0:08 | 1.05E-11 | -0:01 | 4.28E-01 |
| 4 | -22 | -3.13 | 3.90E-13 | -1.62 | 9.20E-06 | -19.28 | 9.99E-25 |
| 17 | 36 | 12.67 | 5.54E-12 | 6.27 | 8.89E-05 | 17.73 | 1.95E-18 |
| 18 | 9 | 3.43 | 2.23E-15 | 1.07 | 3.41E-05 | 4.87 | 1.16E-12 |
| 19 | 960588 | 70046.48 | 3.86E-02 | 10238.70 | 1.12E-02 | 439127.97 | 1.84E-09 |
| 20 | 18 | 1.83 | 1.28E-04 | 3.97 | 1.81E-07 | 8.93 | 8.40E-11 |
| 21 | -51 | -0.60 | 1.63E-01 | -3.86 | 2.05E-03 | -23.65 | 2.13E-12 |
| 22 | 0:39 | 0:32 | 3.21E-06 | 0:08 | 6.36E-02 | -0:39 | 8.13E-01 |
| 23 | -10 | -0.80 | 6.47E-10 | -0.12 | 1.63E-01 | -1.90 | 1.86E-03 |
| 24 | 0.14 | 0.05 | 6.85E-12 | 0.04 | 3.03E-06 | -0.06 | 6.00E-01 |
| 25 | 42 | -0.04 | 5.99E-01 | -0.03 | 1.00E+00 | 5.50 | 4.43E-02 |
| 26 | -1:59 | -0:41 | 1.00E+00 | 0:09 | 3.06E-01 | -0:53 | 1.00E+00 |
| 27 | -7 | -2.45 | 6.12E-21 | -0.53 | 2.16E-03 | -3.12 | 2.58E-09 |
| 28 | 0.017 | 0.00 | 1.00E+00 | 0.00 | 1.00E+00 | 0.05 | 5.31E-02 |
| 29 | 8358 | 3663.24 | 3.18E-08 | 297.07 | 8.51E-02 | 2716.12 | 3.33E-02 |
| 30 | -25 | -10.83 | 1.14E-27 | -10.31 | 1.05E-12 | -16.08 | 7.80E-23 |
| 31 | 17.55 | 4.20 | 1.85E-08 | 3.86 | 1.01E-06 | 3.69 | 4.81E-03 |
| 32 | -32017673 | -1860549.83 | 9.02E-05 | -4885910.97 | 1.47E-04 | 4913650.75 | 6.15E-01 |
| 33 | 2.73 | 2.93 | 2.33E-34 | 2.87 | 1.11E-23 | 2.35 | 5.29E-32 |
| 34 | -32 | -6.10 | 5.04E-06 | -1.74 | 2.23E-02 | -17.23 | 3.87E-08 |
| 38 | 16 | 0.23 | 8.05E-02 | 0.00 | 1.00E+00 | 0.91 | 3.78E-02 |
| 39 | -1.11 | -0.27 | 4.49E-19 | -0.07 | 2.60E-02 | -0.34 | 2.32E-04 |
| 40 | 3:10 | 1:08 | 1.57E-19 | 1:23 | 5.05E-12 | 2:56 | 5.50E-09 |
| 41 | 385 | 155.33 | 1.05E-14 | 191.61 | 3.94E-28 | 263.00 | 5.15E-03 |
| 43 | 32.52 | 4.21 | 2.24E-03 | 18.80 | 1.12E-16 | 68.51 | 2.90E-01 |
| 44 | 0:37 | -0:04 | 7.91E-01 | 0:15 | 8.43E-01 | -1:03 | 9.18E-01 |
| 45 | -10 | -5.20 | 6.70E-28 | -4.82 | 1.29E-15 | -8.17 | 8.49E-24 |
| 47 | -0.0041 | 0.00 | 5.00E-01 | 0.00 | 5.00E-01 | 0.31 | 1.00E+00 |
| 49 | 6000 | 4383.33 | 1.66E-22 | 4923.33 | 1.87E-31 | 5378.90 | 3.81E-17 |
| 50 | 6499 | 330.07 | 2.41E-03 | 137.37 | 2.47E-02 | 12373.77 | 5.07E-03 |
| 51 | -399 | -194.73 | 1.19E-33 | -216.88 | 6.45E-21 | 116.13 | 9.59E-01 |
| 52 | -0:38 | -0:16 | 9.75E-01 | -0:01 | 7.84E-01 | 0:01 | 1.73E-01 |
| 53 | -0.58 | -0.12 | 2.05E-08 | -0.04 | 5.01E-03 | 0.06 | 3.94E-01 |
| 56 | 40 | 14.41 | 9.91E-17 | 1.43 | 8.77E-03 | 17.30 | 4.04E-09 |
| 57 | 2.65 | 0.56 | 1.62E-25 | 0.04 | 5.00E-01 | -0.04 | 5.28E-01 |
| 58 | 149 | 34.73 | 9.28E-04 | 101.37 | 3.14E-06 | 126.28 | 8.14E-02 |
| 59 | 1.88 | 0.20 | 0.00E+00 | 0.25 | 2.33E-04 | 0.33 | 7.72E-02 |
| 60 | 10 | 0.00 | 1.00E+00 | 0.10 | 1.00E+00 | 1.49 | 4.51E-02 |
| 61 | -0:28 | 0:00 | 4.35E-01 | -0:13 | 5.88E-01 | 0:10 | 8.79E-01 |
| 62 | -28 | -7.27 | 9.62E-32 | -3.77 | 3.87E-06 | -22.51 | 4.64E-20 |

**Fig. 4** T-test results of biased answers on GPT models with fact hints

| question_id | Hint Difference | diff | p_value | diff_4o | p_value_4o | diff_35 | p_value_35 |
|---|---|---|---|---|---|---|---|
| 5 | -4 | -2.03 | 0.00E+00 | -2.38 | 3.08E-19 | -5.04 | 1.20E-20 |
| 6 | -150000 | -150000.00 | 0.00E+00 | -150000.00 | 0.00E+00 | -298625.23 | 2.43E-06 |
| 7 | -8 | -7.47 | 3.11E-27 | -7.86 | 8.04E-34 | -6.97 | 1.66E-30 |
| 8 | -16 | -12.69 | 1.18E-27 | -15.33 | 3.67E-49 | -13.80 | 7.03E-19 |
| 9 | 800505 | 800505.00 | 0.00E+00 | 800505.00 | 0.00E+00 | 504840.26 | 3.26E-02 |
| 10 | 3 | 2.70 | 2.41E-21 | 2.86 | 7.15E-32 | 2.13 | 4.94E-14 |
| 11 | 4 | 3.13 | 7.91E-11 | 1.80 | 2.64E-05 | 4.07 | 0.00E+00 |
| 12 | 1700 | 1141.40 | 2.34E-16 | 575.48 | 8.32E-13 | 1126.48 | 6.61E-15 |
| 13 | -2.5 | -2.25 | 4.76E-26 | -2.39 | 2.04E-57 | -1.18 | 3.88E-04 |
| 14 | 6.2 | 6.15 | 0.00E+00 | 5.46 | 6.07E-24 | 4.86 | 1.13E-14 |
| 15 | -0.563 | -0.56 | 0.00E+00 | -0.55 | 9.27E-38 | -0.53 | 1.46E-25 |
| 16 | 4 | 4.30 | 4.42E-20 | 1.23 | 4.51E-02 | 4.00 | 2.17E-11 |

**Fig. 5** T-test results of biased answers on GPT models with expert hints

in set $A$ and set $B$ and measure the confidence of this average difference with $p$-value using t-test analysis. For example, Figs. 4 and 5 show the *t-test* results on different GPT models' responses to questions with different hint types of fact anchoring and expert opinion anchoring, respectively. Here, the column "Hint Difference" shows the difference between the original values in two anchoring hints $H_2$ and $H_3$ for each question, namely $(H_3 - H_2)$. The columns "diff", "diff_4o", and "diff_35" are the average difference values between the two distributions from treatment experiment $A$ and $B$ (e.g., $(B - A)$) on the *GPT4* model, *GPT4o*, and *GPT3.5* respectively. And, "$p$_value", "$p$_value_4o", and "$p$_value_35" are their corresponding t-test confidence values. A smaller p-value (i.e., less than 0.05) often means a higher confidence on the hypothesis that two distributions are different.

In order to measure the level of the influence caused by anchoring hints, we adopt Average Anchoring Index (AI) from [18], which is defined as follows:

$$AI = \frac{\text{median}_{\text{high anchor}} - \text{median}_{\text{low anchor}}}{\text{high anchor} - \text{low anchor}} \tag{1}$$

#### 4.2.1 Observation 1: stronger models show more consistent bias behavior under anchoring hints

From Figs. 4, 6, 7, 8 , we can see that the answers to many questions are biased by the anchoring hints. Specifically, in Figs. 4, 5, most distribution differences (i.e., "diff", "diff_4o", and "diff_35" values in the figure) share the same signs with their corresponding "Hint Difference" values. There are only 11 exception cases (highlighted with red color in the tables) among the total 162 distribution difference values, whose signs are not consistent with the signs of their corresponding "Hint Difference" values. It provides a strong evidence to the anchoring effect of LLMs, although a few $p$-values show that the corresponding differences are not statistically significant enough. It also shows that the answers from stronger models (i.e., *GPT4* and *GPT4o*) are more consistently influenced by anchoring bias. On the contrary, the anchoring effects on *GPT-3.5* are not as stable as that of stronger models (i.e., *GPT4* and *GPT4o*). Among the 11 exceptions, 9 of them come from the answers of *GPT-3.5*. However, this does not mean the weaker model has a high resistance to the influence of anchoring effects. The reason behind of this phenomenon is that a weaker model may introduce a lot of randomness or hallucinations in answer generation, which leads to a high fluctuation of the results and potentially hides the underlying anchoring effects, while stronger models perform consistent behavior.

Similar results can also be observed from Gemini models (refer to Figs. 7 and 8), Claud models (refer to Figs. 10 and 11), and the comparison between *DeepSeek* and *GPTo3* models (refer to Fig. 13). In Figs. 7 and 8, we can see that, *Gemini-2.5 Flash-*

| question_id | Hint Difference | diff | p_value | diff_4o | p_value_4o | diff_35 | p_value_35 |
|---|---|---|---|---|---|---|---|
| 35 | 69 | 20.03 | 1.85E-03 | -1.90 | 5.64E-01 | -4689.24 | 9.98E-01 |
| 36 | -107 | 52.57 | 1.00E+00 | 4.39 | 7.27E-02 | 4142.91 | 7.00E-01 |
| 37 | -25 | -25.00 | 0.00E+00 | -12.43 | 2.25E-05 | -31.90 | 2.71E-12 |
| 42 | 0:27 | -0:06 | 8.13E-01 | -0:30 | 9.82E-01 | -0:11 | 7.86E-01 |
| 46 | -6.83 | -6.18 | 2.02E-22 | 520.63 | 7.49E-01 | -983.23 | 1.47E-01 |
| 48 | 2924 | -1359.53 | 9.34E-01 | -3508.02 | 1.00E+00 | -1710.29 | 9.45E-01 |
| 54 | 159 | 142.19 | 2.50E-40 | 120.51 | 9.12E-10 | 30.17 | 4.89E-02 |
| 55 | 118 | -19.84 | 4.60E-01 | -565.27 | 4.99E-01 | -5684.62 | 9.44E-01 |

**Fig. 6** T-test results of biased answers on GPT models when $H_1$ is irrelevant to the question

| question_id | Hint Difference | diff_gemini-2.5-flash-lite | p_value_gemini-2.5-flash-lite | diff_gemini-2.5-flash | p_value_gemini-2.5-flash |
|---|---|---|---|---|---|
| 1 | 600 | 419.27 | 1.27E-04 | 606.36 | 1.82E-18 |
| 2 | -2.09 | -0.48 | 8.31E-05 | -0.12 | 1.34E-03 |
| 3 | -0:16 | 0:00 | 1.00E+00 | 0.02 | 3.26E-01 |
| 4 | -22 | -0.82 | 8.92E-05 | -3.75 | 1.29E-14 |
| 17 | 36 | 9.93 | 5.65E-18 | 2.54 | 2.84E-03 |
| 18 | 9 | 3.47 | 3.10E-18 | 4.05 | 7.45E-15 |
| 19 | 960588 | 227544.29 | 2.71E-19 | 56419.89 | 9.61E-02 |
| 20 | 18 | 2.45 | 1.14E-04 | 13.00 | 0.00E+00 |
| 21 | -51 | -6.50 | 4.36E-10 | 0.00 | 1.00E+00 |
| 22 | 0:39 | 11.56 | 7.27E-02 | 7.44 | 4.59E-03 |
| 23 | -10 | -1.89 | 1.28E-13 | -0.46 | 1.66E-02 |
| 24 | 0.14 | 0.07 | 1.11E-27 | 0.00 | 1.00E+00 |
| 25 | 42 | -1.31 | 2.60E-04 | 0.00 | 1.00E+00 |
| 26 | -1:59 | 0:00 | 1.00E+00 | 0:00 | 3.26E-01 |
| 27 | -7 | -1.68 | 1.44E-12 | -0.21 | 1.14E-02 |
| 28 | 0.017 | 0.00 | 1.00E+00 | 0.00 | 1.00E+00 |
| 29 | 8358 | 0.00 | 1.00E+00 | 0.00 | 1.00E+00 |
| 30 | -25 | -8.59 | 3.77E-13 | -17.82 | 1.41E-27 |
| 31 | 17.55 | 6.84 | 2.36E-11 | 0.00 | 1.00E+00 |
| 32 | -32017673 | -164012.79 | 3.26E-01 | 0.00 | 1.00E+00 |
| 33 | 2.73 | 3.00 | 0.00E+00 | 3.00 | 0.00E+00 |
| 34 | -32 | -6.02 | 4.17E-06 | -1.79 | 2.82E-08 |
| 38 | 16 | 0.00 | 1.00E+00 | 0.00 | 1.00E+00 |
| 39 | -1.11 | -0.13 | 5.34E-06 | 0.00 | 1.00E+00 |
| 40 | 3:10 | 61.11 | 1.94E-28 | 28.48 | 6.45E-07 |
| 41 | 385 | -41.41 | 5.82E-03 | 192.63 | 2.79E-79 |
| 43 | 32.52 | 4.77 | 1.79E-07 | 2.92 | 4.32E-02 |
| 44 | 0:37 | -4:56 | 4.34E-03 | -0:11 | 2.95E-04 |
| 45 | -10 | -4.96 | 4.33E-40 | -8.00 | 0.00E+00 |
| 47 | -0.0041 | 0.00 | 1.00E+00 | 0.00 | 1.00E+00 |
| 49 | 6000 | 5000.00 | 0.00E+00 | 5750.00 | 6.68E-32 |
| 50 | 6499 | 323.39 | 2.04E-04 | 0.00 | 1.00E+00 |
| 51 | -399 | -170.54 | 3.26E-22 | -196.14 | 9.16E-46 |
| 52 | -0:38 | 0.99 | 1.61E-01 | -8.61 | 1.06E-13 |
| 53 | -0.58 | -0.27 | 1.20E-24 | -0.11 | 1.77E-06 |
| 56 | 40 | 5.86 | 1.47E-13 | 18.84 | 1.88E-08 |
| 57 | 2.65 | 0.29 | 4.55E-05 | -0.02 | 5.14E-01 |
| 58 | 149 | 193.82 | 5.46E-04 | 198.50 | 4.30E-05 |
| 59 | 1.88 | 0.34 | 8.31E-06 | 0.00 | 1.00E+00 |
| 60 | 10 | 2.71 | 1.44E-27 | 0.00 | 1.00E+00 |
| 61 | -0:28 | 21.53 | 5.23E-03 | 1.41 | 1.85E-01 |
| 62 | -28 | -6.68 | 1.41E-15 | -0.50 | 1.61E-01 |

**Fig. 7** T-test results of biased answers on Gemini models with fact hints

| question_id | Hint Difference | diff_gemini-2.5-flash-lite | p_value_gemini-2.5-flash-lite | diff_gemini-2.5-flash | p_value_gemini-2.5-flash |
|---|---|---|---|---|---|
| 5 | -4 | -5.37 | 2.80E-21 | -6.00 | 0.00E+00 |
| 6 | -150000 | -156239.29 | 7.29E-26 | -150000.00 | 0.00E+00 |
| 7 | -8 | -5.64 | 4.77E-29 | -8.00 | 0.00E+00 |
| 8 | -16 | -16.00 | 0.00E+00 | -16.00 | 0.00E+00 |
| 9 | 800505 | 1025420.00 | 1.86E-17 | 800505.00 | 0.00E+00 |
| 10 | 3 | 1.63 | 4.30E-24 | 3.00 | 0.00E+00 |
| 11 | 4 | 1.86 | 4.93E-05 | 4.00 | 0.00E+00 |
| 12 | 1700 | 837.39 | 1.35E-29 | 1700.00 | 0.00E+00 |
| 13 | -2.5 | -1.66 | 6.77E-24 | -2.50 | 0.00E+00 |
| 14 | 6.2 | 5.81 | 6.22E-27 | 6.20 | 0.00E+00 |
| 15 | -0.563 | -0.54 | 1.76E-27 | -0.56 | 1.86E-37 |
| 16 | 4 | 1.36 | 7.06E-02 | 5.71 | 2.97E-13 |

**Fig. 8** T-test results of biased answers on Gemini models with expert hints

*lite* model has more exception cases than *Gemini-2.5 Flash* model, and in Figs. 10 and 12, *Claud Haiku3* model has more exception cases than *Claud Haiku3.5* model. In Fig. 13, compared to *GPT-o3*, a weaker model called *DeepSeek-R1-Qwen-32B* also triggers more exception cases.

#### 4.2.2 Observation 2: models with strong reasoning capability have higher resistance to anchoring biases

It is noteworthy that the difference columns (i.e., *diff_o3* and *diff_gemini-2.5-flash*) for advanced reasoning models such as *GPT-o3* and *Gemini-2.5 Flash* in Figs. 13 and 7 display a significant number of zero values. For any given question, a zero-valued difference between experiments A and B indicates that the model produced identical responses to the question despite exposure to different anchoring hints. In other words, the model's responses are not influenced by different anchoring hints, and more zero-valued differences often indicate its higher resistance to anchoring biases. As depicted in Fig. 13, *GPT-o3* demonstrates 13 zero-valued differences, which is substantially higher than the 3 observed for *DeepSeek-R1-QWen-32B*. This finding aligns with the well known fact that *GPT-o3* exhibits superior reasoning capabilities compared to *DeepSeek-R1-QWen-32B*. Similar trends are evident in Fig. 7, where *Gemini-2.5 Flash* achieves 14 zero-valued differences, whereas *Gemini-2.5 Flash-lite* records only 5 such cases. From the figures, we can find that, for other models not specifically optimized for reasoning tasks, their frequency of zero-valued differences remains significantly lower than those exhibited by *GPT-o3* and *Gemini-2.5 Flash-lite*.

This trend highlights the crucial link between robust reasoning abilities and reduced susceptibility to anchoring effects.

#### 4.2.3 Observation 3: anchoring effects on time-related questions not significant

By looking into cases in Fig. 4, we have an interesting observation that the questions with time-period answers are not easy to be biased. For example, in Fig. 2, we have 7 questions (*no*.3, *no*.22, *no*.40, *no*.26, *no*.44, *no*.52, *no*.61) expecting time-period answers, which are about 17% (7 of 42) of the questions, but they bring about 54% (6 of 11) out of total exceptions. Our hypothesis is that the capability of LLM's time-period prediction is slightly worse than its capability on normal numbers. A similar conclusion can also be obtained from other two series of models in Figs. 7 and 10.

#### 4.2.4 Observation 4: LLMs are much easier to be biased by expert anchors

Figure 5 shows that the answers from GPT models are biased by expert anchors with high confidence. For example, there is no exception in the answers of 12 questions whose anchoring hints come from a PTF (Play The Future) expert. Furthermore, their corresponding *p*-values are also very small, which means that the test has a very high level of confidence. A similar phenomenon can be found in Figs. 8, 11, and 14 for both Gemini, Claud models, and DeepSeek models. These results show that all models including strong and weak models tend to strongly adherent to the expert hints from PTF.

#### 4.2.5 Observation 5: anchoring effects are not significant when given irrelevant hints

Figures 6, 9, 12, and 15 illustrate a high rate of exception cases in which all GPT models, Gemini models, Claud models, and DeepSeek models do not exhibit consistent behaviors when the $H_1$ hint lacks relevant information for the question.

A plausible explanation is as follows: Under the prompt condition $H_1$, where the input includes information that is irrelevant to the task, large language models tend to produce a broader distribution of answers. In other words, their outputs become more

| question_id | Hint Difference | diff_claude-haiku3 | p_value_claude-haiku3 | diff_claude_haiku35 | p_value_claude_haiku35 |
|---|---|---|---|---|---|
| 1 | 600 | 189.29 | 4.85E-05 | 300.00 | 0.00E+00 |
| 2 | -2.09 | -0.95 | 2.17E-49 | -0.52 | 5.67E-20 |
| 3 | -0:16 | 0.00 | 1.00E+00 | 0.00 | 1.00E+00 |
| 4 | -22 | -5.46 | 1.02E-20 | -8.00 | 0.00E+00 |
| 17 | 36 | 18.86 | 1.36E-20 | 12.04 | 2.77E-17 |
| 18 | 9 | 4.61 | 3.20E-24 | 0.14 | 1.61E-01 |
| 19 | 960588 | 955093.36 | 5.43E-47 | 7054.29 | 3.81E-02 |
| 20 | 18 | -7.50 | 2.09E-10 | 4.71 | 2.71E-15 |
| 21 | -51 | -22.25 | 8.71E-17 | -14.36 | 1.43E-16 |
| 22 | 0:39 | 39.23 | 5.57E-05 | 48.68 | 8.29E-12 |
| 23 | -10 | -4.00 | 0.00E+00 | -0.21 | 3.11E-02 |
| 24 | 0.14 | 0.07 | 4.05E-36 | 0.06 | 6.37E-13 |
| 25 | 42 | 20.75 | 7.63E-36 | -0.11 | 8.30E-02 |
| 26 | -1:59 | -60.00 | 0.00E+00 | -19.35 | 1.29E-09 |
| 27 | -7 | -2.11 | 3.68E-27 | -1.50 | 1.57E-08 |
| 28 | 0.017 | 0.00 | 1.00E+00 | 0.00 | 1.00E+00 |
| 29 | 8358 | 8358.00 | 0.00E+00 | 69.93 | 3.26E-01 |
| 30 | -25 | -11.18 | 1.54E-30 | -20.00 | 0.00E+00 |
| 31 | 17.55 | 1.54 | 8.38E-02 | 3.41 | 4.82E-18 |
| 32 | -32017673 | -14008162.11 | 1.47E-20 | -7012573.50 | 1.12E-14 |
| 33 | 2.73 | 3.14 | 2.36E-27 | 2.04 | 1.12E-29 |
| 34 | -32 | -8.14 | 1.96E-07 | -0.29 | 4.32E-02 |
| 38 | 16 | 2.29 | 2.11E-18 | 0.00 | 1.00E+00 |
| 39 | -1.11 | -0.29 | 3.36E-08 | -0.24 | 0.00E+00 |
| 40 | 3:10 | 60.00 | 0.00E+00 | 60.00 | 0.00E+00 |
| 41 | 385 | 133.04 | 7.55E-03 | 188.82 | 3.13E-43 |
| 43 | 32.52 | 2.55 | 4.30E-02 | 11.18 | 3.10E-23 |
| 44 | 0:37 | 23.04 | 5.46E-20 | 9.54 | 1.25E-04 |
| 45 | -10 | -5.00 | 0.00E+00 | -7.86 | 1.61E-33 |
| 47 | -0.0041 | 0.00 | 1.92E-56 | 0.00 | 1.00E+00 |
| 49 | 6000 | 3642.86 | 2.10E-15 | 3750.00 | 5.31E-18 |
| 50 | 6499 | 3514.54 | 1.15E-30 | 510.39 | 1.02E-12 |
| 51 | -399 | -352.32 | 1.53E-17 | -203.79 | 3.79E-44 |
| 52 | -0:38 | 0.00 | 1.00E+00 | -0.90 | 4.15E-01 |
| 53 | -0.58 | -0.34 | 7.98E-13 | -0.18 | 4.25E-42 |
| 56 | 40 | 11.61 | 1.23E-20 | 6.86 | 1.04E-08 |
| 57 | 2.65 | -0.25 | 3.65E-01 | 0.00 | 1.00E+00 |
| 58 | 149 | 126.93 | 1.46E-28 | 84.64 | 9.06E-12 |
| 59 | 1.88 | -0.07 | 1.36E-01 | 0.00 | 1.00E+00 |
| 60 | 10 | 2.18 | 8.48E-20 | 0.00 | 1.00E+00 |
| 61 | -0:28 | 0.00 | 1.00E+00 | -1.54 | 6.56E-01 |
| 62 | -28 | -22.61 | 5.36E-27 | -0.61 | 8.76E-02 |

**Fig. 9** T-test results of biased answers on Gemini models when $H_1$ is irrelevant to the question

dispersed and the variance across samples increases. This suggests that irrelevant hint inflates the model's uncertainty. At first glance, this might seem to conflict with the previous literature showing that irrelevant information systematically influences decisions [19]. On closer inspection, however, we can see that there is no contradiction. Both findings indicate both human and LLM's susceptibility to irrelevant cues; they simply differ in how that susceptibility manifests because the experiment setups are different. In classic human experiments, irrelevant cues are used as anchoring hints that often introduce a directional bias–such as an anchoring effect–that consistently shifts judgments in a particular direction. By contrast, in our experiment, the irrelevant content $H_1$ is used as a distraction information from anchoring hints of $H_2$ and $H_3$, which tends to destabilize the inference process of large language models. As a result, it increases dispersion in question answering, and the variability induced by irrelevant information can mask or outweigh any subtle underlying anchoring effect caused by $H_2$ or $H_3$.

## 5 Mitigation strategies

As cognitive bias may largely influence the correctness of decision making, it is important to know whether we can mitigate the anchoring bias of LLMs during inference. In this section, we list several potential strategies and test whether they can mitigate anchoring bias effectively.

Here is a list of strategies we try to study in this paper:

- **Chain-of-Thought (CoT)**: it is well known that chain-of-thought can significantly improve reasoning accuracies of LLM models [20]. It is reasonable to ask whether CoT can mitigate anchoring bias of LLMs.
- **Thoughts of Principles**: some previous studies [21] have shown if we ask LLM to generate principles before reasoning, the overall reasoning accuracy can be boosted. In this section, we test whether the thoughts-of-principle strategy (denoted as PoT in our experiments) can reduce the influence of anchoring bias.
- **Ignore Anchor Hint**: as we know, the bias is caused by anchor hints. In this strategy, similar to [7], we try to mitigate the anchoring effect by explicitly asking LLMs to ignore the anchor hints.
- **Reflection**: reflection is another widely used prompting strategy that can boost reasoning accuracy of LLMs. It lets a LLM mimic human's reflective thinking to improve reasoning accuracy through self-correcting its reasoning steps iteratively. In this section, we also check whether we can use reflection prompting to mitigate anchoring bias.
- **Both-Anchor**: In human decision-making practices, a practical strategy to mitigate the impact of anchoring bias is to gather information from as many angles as possible to serve decision-making. This prevents our cognition from being anchored to individual pieces of information, thereby allowing for a more comprehensive judgment [22]. To simulate multi-angle information, in this study, we attempted to include both $H_2$ and $H_3$ in the prompts to LLM to simulate a situation of more comprehensive prompt information.

In our experiments, we modify our prompt template to integrate these mitigation strategies. For example, prompt *b* in Appendix is our prompt template for the "ignore" strategy, where we ask the LLM to ignore the anchor hint by adding an instruction "The hint part contains an answer from a PTF expert and please **ignore** it when you are answering the question" into the prompt. Then, we test these prompts using *GPT4* and *GPT4o* on our "expert" anchoring questions. We use the "expert" anchoring questions because the anchoring effects are very significant for these questions. [23].

Figures 16 and 17 are our results obtained from *GPT4o* and *GPT4* respectively. Here, diff_cot, diff_pot, diff_ign, diff_ref, and p_value_cot, p_value_pot, p_value_ ign, p_value _ref are the differences and p-values under different strategies including Chain of Thought (CoT), Principle of Thought (PoT), Ignore Strategy (Ign), and Refection Strategy (Ref) respectively. From the figures, we can see that the answers from models with hints of $H_2$ and $H_3$ are still consistently influenced by the anchoring hints, and their corresponding $p$-values are very small. By comparing with Fig. 5, we do not find any significant improvements. It seems that all these strategies are not effective enough to mitigate the anchoring bias for these questions. Similarly, Simmons et al. also observed that it is difficult to completely eliminate anchoring bias from their human experiments [23].

Different from other strategies, our Both-Anchor strategy does not try to directly eliminate anchoring effect. Instead, it includes both $H_2$ and $H_3$ in the prompt to prevent the LLMs from being anchored to an individual hint. We conducted experiments on questions with expert hints using *GPT4*, and obtained the results shown in Fig. 18 (noted as "Both_anchor" column). In the figure, we observed that when the values of the $H_2$ and $H_3$ prompts are on either side of the bias-free benchmark value (e.g., refer to the questions no.5, no.7, no.8, no.10, no.12, no.13, no.14, and no.15), the answers of *GPT4* under the Both-Anchor strategy are very close to the bias-free benchmark value. On the contrary, the answers of *GPT4* to the remaining questions still deviate significantly from the benchmark value. This phenomenon suggests that the advice from human psychology [22] can be applicable to mitigate the impact of anchoring bias in LLMs in practical applications to certain extent, although it does not try to eliminate anchoring bias from a single anchoring hint.

## 6 Discussion

### 6.1 Comparison to human anchoring effects

It is interesting to conduct a comparative study examining the anchoring bias present in large language models (LLMs) alongside the anchoring effects observed in humans. We compared our results from LLMs to the results from the human user study in [8].

*First*, in the human user study on the same dataset from [8], the human responses often have non-zero variations and the authors noticed loyal users in the control group make predictions that resemble the median predictions more closely than casual users. In our experiments, we have a similar observation (refer to Sect. 4.1) that stronger models (i.e., *GPT4*, *GPT4o*) have smaller variations than the weaker model (i.e., *GPT3.5*). In other words, whether it's a human or a model, there is a common rule: as the ability improves, the variance of prediction becomes smaller and the accuracy becomes higher.

*Second*, the average anchoring index (AI [18]) of *GPT4*, *Claud3.5-haiku*, and *Gemini2.5-flash* are about 0.45, 0.39, and 0.46 in our experiment respectively, it is different from 0.61 in the human study [8]. In other words, the level of influence created by anchoring hints on LLMs is smaller than that of humans. It seems that human is easier to be influenced by anchoring hints.

*Third*, although the analysis method is different from that in the human user study [8], from Figs. 4, 7 and 10, we also can have the same conclusion that a larger stimu-

| question_id | Hint Difference | diff_claude-haiku3 | p_value_claude-haiku3 | diff_claude_haiku35 | p_value_claude_haiku35 |
|---|---|---|---|---|---|
| 5 | -4 | -4.04 | 2.62E-36 | -5.50 | 4.05E-26 |
| 6 | -150000 | -151785.71 | 1.86E-36 | -97321.43 | 6.76E-21 |
| 7 | -8 | -4.36 | 4.22E-22 | -6.00 | 0.00E+00 |
| 8 | -16 | -16.00 | 0.00E+00 | -11.86 | 4.66E-34 |
| 9 | 800505 | 535323.39 | 2.37E-15 | 873204.61 | 4.35E-18 |
| 10 | 3 | 2.95 | 1.61E-33 | 1.48 | 4.28E-24 |
| 11 | 4 | 2.00 | 0.00E+00 | 2.00 | 0.00E+00 |
| 12 | 1700 | 1700.00 | 0.00E+00 | 850.00 | 0.00E+00 |
| 13 | -2.5 | -2.20 | 1.88E-26 | -2.36 | 1.44E-27 |
| 14 | 6.2 | 3.85 | 5.56E-30 | 4.61 | 2.04E-40 |
| 15 | -0.563 | -0.41 | 1.41E-24 | -0.30 | 2.46E-34 |
| 16 | 4 | 4.39 | 3.80E-14 | 4.00 | 0.00E+00 |

**Fig. 10** T-test results of biased answers on Claud models with fact hints

| question_id | Hint Difference | diff_ds | p_value_ds | diff_o3 | p_value_o3 |
|---|---|---|---|---|---|
| 1 | 600 | 512.00 | 2.55E-03 | 546.43 | 1.80E-29 |
| 2 | -2.09 | -0.52 | 5.17E-05 | -0.52 | 5.08E-10 |
| 3 | -0:16 | -4.12 | 4.71E-04 | 0.00 | 1.00E+00 |
| 4 | -22 | -5.46 | 8.05E-09 | -11.29 | 1.43E-18 |
| 17 | 36 | 4.21 | 3.28E-04 | 1.93 | 8.30E-02 |
| 18 | 9 | 5.75 | 5.53E-31 | 4.64 | 1.31E-25 |
| 19 | 960588 | 309980.08 | 2.90E-05 | 229317.79 | 2.54E-04 |
| 20 | 18 | 5.32 | 9.77E-21 | 2.46 | 2.15E-07 |
| 21 | -51 | -2.54 | 6.93E-03 | -11.39 | 1.37E-11 |
| 22 | 0:39 | -0.03 | 9.96E-01 | 60.00 | 0.00E+00 |
| 23 | -10 | -0.18 | 1.34E-01 | 0.00 | 1.00E+00 |
| 24 | 0.14 | 0.07 | 2.53E-10 | 0.06 | 6.82E-13 |
| 25 | 42 | -0.07 | 3.27E-01 | 0.00 | 1.00E+00 |
| 26 | -1:59 | 6.47 | 4.90E-01 | 0.00 | 1.00E+00 |
| 27 | -7 | -1.61 | 1.21E-07 | -3.93 | 1.67E-15 |
| 28 | 0.017 | 0.00 | 1.00E+00 | 0.00 | 1.00E+00 |
| 29 | 8358 | 8423.06 | 3.26E-03 | 7550.07 | 3.27E-03 |
| 30 | -25 | -16.32 | 9.50E-22 | -17.18 | 4.85E-22 |
| 31 | 17.55 | 4.08 | 1.98E-05 | 4.71 | 2.96E-14 |
| 32 | -32017673 | -15463551.49 | 2.05E-23 | -5412421.57 | 3.02E-09 |
| 33 | 2.73 | 1.96 | 3.75E-02 | 3.00 | 0.00E+00 |
| 34 | -32 | -0.50 | 5.75E-03 | -0.07 | 3.26E-01 |
| 38 | 16 | 0.00 | 1.00E+00 | 0.00 | 1.00E+00 |
| 39 | -1.11 | -0.07 | 1.17E-02 | -0.04 | 2.24E-02 |
| 40 | 3:10 | 92.38 | 2.78E-10 | 55.71 | 5.30E-17 |
| 41 | 385 | 282.93 | 8.13E-16 | 103.57 | 8.10E-08 |
| 43 | 32.52 | 22.47 | 1.39E-22 | 12.44 | 9.20E-22 |
| 44 | 0:37 | -30.79 | 2.60E-01 | -12.86 | 5.61E-02 |
| 45 | -10 | -5.36 | 2.30E-19 | -6.68 | 3.73E-20 |
| 47 | -0.0041 | 0.00 | 6.37E-02 | 0.00 | 1.00E+00 |
| 49 | 6000 | 4272.73 | 2.16E-08 | 5000.00 | 0.00E+00 |
| 50 | 6499 | 2543.36 | 1.11E-15 | 101.25 | 3.26E-01 |
| 51 | -399 | -237.00 | 3.01E-19 | -261.11 | 1.75E-07 |
| 52 | -0:38 | -3.03 | 1.52E-01 | 0.00 | 1.00E+00 |
| 53 | -0.58 | -0.12 | 6.75E-05 | -0.08 | 1.18E-04 |
| 56 | 40 | 4.14 | 7.32E-05 | 0.00 | 1.00E+00 |
| 57 | 2.65 | 0.10 | 1.13E-01 | 0.00 | 1.00E+00 |
| 58 | 149 | 144.71 | 4.56E-04 | 81.50 | 5.15E-14 |
| 59 | 1.88 | 0.34 | 1.28E-03 | 0.06 | 6.19E-04 |
| 60 | 10 | 0.36 | 2.24E-02 | 0.00 | 1.00E+00 |
| 61 | -0:28 | 11.61 | 2.28E-01 | 0.00 | 1.00E+00 |
| 62 | -28 | -0.18 | 3.26E-01 | 0.00 | 1.00E+00 |

**Fig. 11** T-test results of biased answers on Claud models with expert hints

| question_id | Hint Difference | diff_claude-haiku3 | p_value_claude-haiku3 | diff_claude_haiku35 | p_value_claude_haiku35 |
|---|---|---|---|---|---|
| 35 | 69.00 | 69.00 | 0.00E+00 | 43.89 | 6.80E-23 |
| 36 | -107.00 | 26.25 | 5.06E-01 | -13.96 | 5.20E-01 |
| 37 | -25.00 | -25.00 | 0.00E+00 | -31.32 | 3.91E-33 |
| 42 | 0.02 | 2.93 | 6.36E-01 | -42.64 | 1.64E-11 |
| 46 | -6.83 | -0.35 | 7.63E-01 | 1.04 | 2.71E-01 |
| 48 | 2924.00 | -255.11 | 7.01E-01 | 1066.07 | 3.53E-02 |
| 54 | 159.00 | 101.41 | 6.79E-09 | 146.52 | 1.07E-33 |
| 55 | 118.00 | -84.22 | 8.31E-03 | -249.68 | 1.00E-23 |

**Fig. 12** T-test results of biased answers on Claud models when $H_1$ is irrelevant to the question

| question_id | Hint Difference | diff_ds | p_value_ds | diff_o3 | p_value_o3 |
|---|---|---|---|---|---|
| 5 | -4 | -5.29 | 9.46E-22 | -6.00 | 0.00E+00 |
| 6 | -150000 | -150000.00 | 0.00E+00 | -150000.00 | 0.00E+00 |
| 7 | -8 | -7.36 | 3.05E-35 | -8.00 | 0.00E+00 |
| 8 | -16 | -15.14 | 5.10E-42 | -16.00 | 0.00E+00 |
| 9 | 800505 | 836440.89 | 2.11E-19 | 800505.00 | 0.00E+00 |
| 10 | 3 | 2.49 | 5.89E-28 | 3.00 | 0.00E+00 |
| 11 | 4 | 3.79 | 2.59E-13 | 3.93 | 5.27E-21 |
| 12 | 1700 | 1087.24 | 8.96E-16 | 1608.93 | 6.18E-23 |
| 13 | -2.5 | -2.13 | 1.47E-31 | -2.50 | 0.00E+00 |
| 14 | 6.2 | 6.00 | 2.79E-45 | 6.20 | 0.00E+00 |
| 15 | -0.563 | -0.48 | 2.37E-30 | -0.56 | 2.53E-141 |
| 16 | 4 | 2.85 | 3.01E-04 | 4.00 | 0.00E+00 |

**Fig. 13** T-test results of Deepseek (i.e., ds) & O3 models with fact hints

| question_id | Hint Difference | diff_gemini-2.5-flash-lite | p_value_gemini-2.5-flash-lite | diff_gemini-2.5-flash | p_value_gemini-2.5-flash |
|---|---|---|---|---|---|
| 35 | 69 | -912.80 | 7.75E-02 | -249.86 | 4.11E-01 |
| 36 | -107 | 265.36 | 1.81E-06 | 187.36 | 4.94E-02 |
| 37 | -25 | -26.09 | 1.42E-26 | -25.00 | 0.00E+00 |
| 42 | 0.01875 | 26.45 | 4.94E-01 | 4.68 | 1.27E-01 |
| 46 | -6.83 | -8.70 | 2.63E-01 | -23.39 | 5.55E-05 |
| 48 | 2924 | -2334.29 | 3.95E-08 | 2001.96 | 7.25E-17 |
| 54 | 159 | -9.99 | 6.91E-01 | 45.69 | 1.85E-14 |
| 55 | 118 | -709.23 | 7.41E-07 | 114.75 | 3.91E-24 |

**Fig. 14** T-test results of Deepseek (i.e., ds) & O3 models with expert hints

| question_id | Hint Difference | diff_ds | p_value_ds | diff_o3 | p_value_o3 |
|---|---|---|---|---|---|
| 35 | 69 | -454.29 | 3.16E-01 | 32.86 | 3.93E-05 |
| 36 | -107 | 35.93 | 1.92E-01 | -23.43 | 1.08E-02 |
| 37 | -25 | 3.55 | 1.57E-13 | -25.00 | 0.00E+00 |
| 42 | 0:27 | -45.37 | 2.53E-03 | -60.00 | 0.00E+00 |
| 46 | -6.83 | 11.07 | 2.07E-01 | -33.37 | 1.12E-02 |
| 48 | 2924 | 1462.32 | 4.18E-02 | -3925.11 | 2.70E-04 |
| 54 | 159 | -54.72 | 8.45E-08 | 80.14 | 2.55E-04 |
| 55 | 118 | -530.61 | 3.54E-03 | 118.00 | 0.00E+00 |

**Fig. 15** T-test results of Deepseek (i.e., ds) & O3 models with irrelevant hints

| question_id | Hint Difference | diff_cot | p_value_cot | diff_pot | p_value_pot | diff_ign | p_value_ign | diff_ref | p_value_ref |
|---|---|---|---|---|---|---|---|---|---|
| 5 | -4 | -4.79 | 4.15E-34 | -3.64 | 1.16E-13 | -4.32 | 4.24E-19 | -4.5 | 1.55E-21 |
| 6 | -150000 | -138137.64 | 4.58E-11 | -133494.9 | 2.81E-12 | -160696.32 | 6.45E-15 | -89414.18 | 1.08E-04 |
| 7 | -8 | -7.86 | 8.04E-34 | -7.71 | 2.29E-27 | -8 | 0.00E+00 | -7.71 | 4.93E-30 |
| 8 | -16 | -15.86 | 9.30E-38 | -14.82 | 2.33E-32 | -15.57 | 3.20E-49 | -14.86 | 1.24E-26 |
| 9 | 800505 | 762334.33 | 1.15E-07 | 1101009.47 | 7.55E-07 | 772272.68 | 2.20E-21 | 727922.18 | 2.43E-16 |
| 10 | 3 | 2.78 | 1.49E-23 | 2.8 | 4.62E-31 | 2.98 | 1.53E-42 | 2.47 | 4.54E-29 |
| 11 | 4 | 2 | 8.99E-06 | 2.33 | 9.28E-04 | 3.48 | 2.24E-13 | 3.06 | 1.21E-09 |
| 12 | 1700 | 895.93 | 2.52E-18 | 590.91 | 9.75E-14 | 268.25 | 5.02E-05 | 674.32 | 2.05E-15 |
| 13 | -2.5 | -2.49 | 9.45E-51 | -2.08 | 1.69E-23 | -2.49 | 2.55E-43 | -2.28 | 1.98E-31 |
| 14 | 6.2 | 5.74 | 1.05E-40 | 4.14 | 4.07E-20 | 5.98 | 5.42E-31 | 5.09 | 3.38E-33 |
| 15 | -0.563 | -0.55 | 4.50E-49 | -0.53 | 8.81E-45 | -0.51 | 4.76E-35 | -0.5 | 3.50E-35 |
| 16 | 4 | 3.21 | 4.34E-08 | 3.43 | 1.48E-07 | 2.5 | 9.52E-04 | 1.79 | 1.10E-03 |

**Fig. 16** T-test results of biased answers under different mitigation strategies with *GPT4o*

lus value (refer to the Hint difference column that is the value difference between hints $H_2$ and $H_3$) often leads to a larger anchoring response.

*Fourth*, [8] reports that human users' responses to the questions containing PTF-prediction values are consistently larger than that to the standard questions. In our experiment, the results also show that the responses of LLMs are strongly adherent to

| question_id | Hint Difference | diff_cot | p_value_cot | diff_pot | p_value_pot | diff_ign | p_value_ign | diff_ref | p_value_ref |
|---|---|---|---|---|---|---|---|---|---|
| 5 | -4 | -4.79 | 3.32E-40 | -5.85 | 2.75E-14 | -5.47 | 1.20E-42 | -4.59 | 7.92E-24 |
| 6 | -150000 | -141703.84 | 4.02E-21 | -173626.26 | 1.67E-11 | -136930.43 | 5.76E-13 | -146348.81 | 2.59E-31 |
| 7 | -8 | -7.32 | 3.31E-27 | -7.60 | 4.97E-26 | -8.00 | 0 | -7.76 | 8.62E-29 |
| 8 | -16 | -15.00 | 1.79E-35 | -15.50 | 1.66E-31 | -13.93 | 3.94E-26 | -14.96 | 7.50E-23 |
| 9 | 800505 | 930794.75 | 1.48E-08 | 510600.82 | 0.056111206 | 799032.89 | 4.24E-19 | 783942.80 | 1.45E-38 |
| 10 | 3 | 2.26 | 9.74E-21 | 2.60 | 2.28E-26 | 1.91 | 4.00E-12 | 1.91 | 4.39E-10 |
| 11 | 4 | 3.32 | 7.66E-11 | 3.23 | 4.54E-12 | 3.98 | 1.51E-18 | 2.98 | 1.16E-08 |
| 12 | 1700 | 1057.71 | 2.59E-25 | 1058.70 | 1.89E-21 | 702.73 | 1.51E-10 | 1046.01 | 1.84E-19 |
| 13 | -2.5 | -2.15 | 2.29E-36 | -2.12 | 3.98E-25 | -2.29 | 5.81E-26 | -2.05 | 8.10E-20 |
| 14 | 6.2 | 5.43 | 1.17E-31 | 5.75 | 5.85E-36 | 4.51 | 1.05E-16 | 4.59 | 3.76E-09 |
| 15 | -0.563 | -0.50 | 5.53E-35 | -0.53 | 1.74E-33 | -0.56 | 1.75E-51 | -0.54 | 1.43E-34 |
| 16 | 4 | 2.81 | 1.98E-06 | 3.51 | 2.24E-09 | 3.48 | 7.52E-13 | 3.81 | 1.49E-08 |

**Fig. 17** T-test results of biased answers under different mitigation strategies with *GPT4*

| Question_id | No_anchor | Anchor_H2_CoT | Anchor_H3_CoT | Both_anchor |
|---|---|---|---|---|
| 5 | 10.00 | 13.52 | 8.73 | 10.07 |
| 6 | 234700.00 | 522600.00 | 381525.14 | 453543.33 |
| 7 | 78.00 | 81.93 | 74.70 | 78.00 |
| 8 | 68.00 | 75.73 | 60.90 | 68.53 |
| 9 | 4206205.00 | 5307623.07 | 6230228.00 | 5692238.00 |
| 10 | 100.76 | 100.10 | 102.36 | 100.93 |
| 11 | 16.00 | 19.50 | 22.76 | 22.00 |
| 12 | 3078.00 | 2695.03 | 3737.55 | 3078.00 |
| 13 | 59.06 | 60.25 | 58.11 | 58.74 |
| 14 | 166.61 | 164.23 | 169.59 | 166.58 |
| 15 | 62.38 | 62.64 | 62.14 | 62.38 |
| 16 | 6.00 | 9.00 | 11.70 | 10.00 |

**Fig. 18** The average answer values obtained from *GPT4* from the CoT strategy, the Both-Anchor strategy, and without anchoring hints ($No\_anchor$ column)

the anchored hints of PTF-prediction. It seems that both human users and LLMs are easy to be influenced by expert-opinions.

*Fifth*, previous work [10] suggests that human anchoring bias is remarkably difficult to overcome. It persists despite people's expertise and experience [24], regardless of incentives to avoid them or explicit warnings of their nature [24, 25]. These findings of human anchoring effect are consistent with our observations during the mitigation experiments, where many of the proposed mitigation strategies were ineffective. In our exploration, the only effective approach was to equip the language model with comprehensive background knowledge, enabling it to make informed decisions and avoid anchoring to a single information source.

## 6.2 Implications to LLM users

Trust in the Model's "Truthfulness" is Fragile: The study highlights that models can be seriously biased by some anchoring information. Users of LLMs need to be careful about this model behavior when they are using LLMs for their day-to-day business tasks.

The Source of the Anchor Matters: The study found that LLMs are significantly more susceptible to anchoring bias when the anchor hint is attributed to a perceived "expert". This suggests that LLMs don't just passively absorb information; they can weigh the credibility of a source and tend to blindly rely on the input from a perceived “expert”.

Mitigation Is Not a Simple Prompting Fix: The research demonstrates that common-sense mitigation techniques like Chain-of-Thought, telling the model to “ignore” the hint, or encouraging reflection are largely ineffective. This implies that debiasing is not a simple prompt engineering problem but a more fundamental issue requiring deeper algorithmic solutions (e.g., collecting comprehensive grounding information) that go beyond current prompting strategies. It should be incorporated into the recent emergence of context engineering framework [26].

## 7 Conclusion

In conclusion, our study shows that anchoring bias is widely prevalent in large language models (LLMs) and poses a significant challenge to their reliability and fairness. In particular, strong models consistently show their vulnerability to the bias of the anchoring effect. They are very easy to be influenced by the “expert” opinions presented in the prompt, and we cannot correct their behavior even when we explicitly ask them not to follow the expert opinions. At the same time, we tried various strategies, including Chain-of-Thought, Thoughts of Principles, Ignoring Anchor Hints, and Reflection, to mitigate anchoring bias in LLMs. Despite the promise application of these strategies in many reasoning tasks, our empirical results on *GPT4* and *GPT4o* indicate that none of these simple mitigating strategies can effectively reduce the anchoring bias in responses to “expert” anchoring questions. This finding underscores the complexity of cognitive biases in AI and highlights the need for more research to develop more robust techniques to minimize bias. Furthermore, our study also shows that, although it is impossible to completely eliminate anchoring effects, in practice, one can collect hints from comprehensive angles to mitigate the impact of anchoring bias by preventing the LLMs from being anchored to individual pieces of information. Our research findings offer valuable insights for scholars studying algorithmic fairness, LLM developers, and policymakers.

All prompt templates, source code, and LLM answers can be found from the URL: https://github.com/JiaxuLou/LLM_Bias.

# Appendix

## Prompt template with anchoring hints

```
< $im_start$ >System
You are a Copilot, a thoughtful and intelligent assistant to help answer a user question. In addition to the question description, there is at least one hint that may or may not be relevant to the user question. You must answer the question using an educated guess based on the provided information. Each answer is a number following the proper unit mentioned in the user question.
Your answer **must** uses the following format.
### Output format:
{
"number": ...,
"unit": ...
}
< $im_end$ >
< $im_start$ >assistant
OK I'm an assistant and I will be helping users to give an educated guess for each user question.
< $im_end$ >
< $im_start$ >User
Good! Here are the questions and hints.
### Question
user_question
### Here are some hints
hint_1
hint_2
Please give your answer!
< $im_end$ >
< $im_start$ >assistant
OK, based on the provided information, I give the following answer. The answer to the question is:
```

## An example of prompt and answer from GPT4

### The question

```
< $im_start$ >System
    You are a Copilot, a thoughtful and intelligent assistant to help answer a user question. In addition to the question description, there is at least one hint that may or may not be relevant to the user question. You must answer the question using an educated guess based on the provided information. Each answer is a number following the proper unit mentioned in the user question.
    Your answer **must** uses the following format.
    ### Output format:
    {
    ”number”: ...,
    ”unit”: ...
    }
    < $im_end$ >
    < $im_start$ >assistant
    OK I’m an assistant and I’ll be helping users to give an educated guess for each user question.
    < $im_end$ >
    < $im_start$ >User
    Good! Here are the questions and hints.
    ### Question
    Apple’s Worldwide Developers Conference will take place next week in San Jose where exciting new software and devices may be presented! What will the starting price be in US $ of the device that is announced last during Apple’s WWDC 2018?
    ### Here are some hints
    Current starting price of Apple’s iPhone 8: 699 USD.
    Current starting price of Apple’s Watch Series 3 (Cellular): 399 USD
    Please give your answer!
    < $im_end$ >
    < $im_start$ >assistant
    OK, based on the provided information, I give the following answer. The answer to the Question is:
```

## Answer from GPT4

{
"number": 499,
"unit": "USD"
}

< $im_start$ >System

You are a Copilot, a thoughtful and intelligent assistant to help answer a user question. In addition to the question description, there is at least one hints that may or may not be relevant to the user question. **The hint part contains an answer from a PTF expert, and please **ignore** it when answering the question.** You must answer the question using an educated guess based on the provided information. Please first illustrate your principles how you obtain the answer and then give your answer. Each answer is a number following the proper unit mentioned in the user question.

Your answer [must] uses the following format.

### Output format:

{
"principles": ...,
"number": ...,
"unit": ...
}

< $im_end$ >

< $im_start$ >assistant

OK I'm an assistant and I'll be helping users to give an educated guess for each user question.

< $im_end$ >

< $im_start$ >User

Good! Here is the question and hints.

### Question

user_question

### Hints

hint_1

hint_2

< $im_end$ >

< $im_start$ >assistant

**OK, based on the provided information, I give the answer step by step.**

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**Data availibility** The data in this paper can be found at the following website: https://github.com/JiaxuLou/LLM_Bias

## Declarations

**Conflict of interest** On behalf of all authors, the corresponding author states that there is no Conflict of interest.



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